

how it is heard on loudspeakers and headphones in real life. In mono sound, all the audio signals are mixed together and routed through a single audio channel. So even if the mono system has multiple loudspeakers, and even multiple widely separated loudspeakers, the key is that the signal does not contain any level and arrival time/phase information. Hence, the volume and sound levels are uniformly heard from all vantage points. However, surround sound technologies take advantage of the presence of multiple speakers in a loudspeaker arrangement and route different channels to the corresponding speaker. So, if the original audio was encoded for reproduction in a 5 channel system, we'd require 5 speakers placed at different points, to hear the audio as the sound engineer had intended it to be heard.

So after we have a mono system, which is basically one speaker, and stereo systems, which have two channels to route the audio signals, we finally come to Surround Sound, also known as multichannel sound, which is created by at least four and up to seven independent audio channels and speakers placed in front of and behind the listener. As the name suggests, the reason behind recording and encoding audio in multiple channels is to 'surround' the listener in sound and provide a realistic experience. In addition, each unique channel is designed to reproduce sound effects that are spatial — which means that the channels will operate at cohesively to create a sound experience that appears to originate from different locations relative to you, the listener. The objective that is achieved with this is that the sound is reproduced in a way that it was originally intended to be heard by the studio. This is why we mentioned that channel configuration and surround sound become important in home theatre systems and concert hall sound systems.

Only commercial surround sound media, including, but not limited to DVDs, and HDTV broadcasts encoded as compressed Dolby Digital and DTS, and lossless audio such as DTS HD Master Audio and Dolby TrueHD on Blu-ray Disc and HD DVD, which are identical to the studio master, are capable of reproducing surround sound formats, when coupled with the appropriate speaker system. Other commercial formats include DVD-Audio (DVD-A) and Super Audio CD (SACD) formats, and MP3 Surround.

Anyone looking to set up their own system would be looking to understand the specific notation that accompanies these systems. The key lies in understanding the placement of the speakers through the room. The current market has many alternative setups available for a surround sound experience, but the 3-2 (3 front, 2 back speakers and a Low Frequency